

Mathematics 101

Integrals, Vectors, Sequences, and Trigonometry

Allowed aids: Calculator, writing materials, and formula sheet.

Task 1

Compute the integrals.

a) $\int_0^{\pi} x \cos x \, dx$

Vis utregningene dine i denne boksen

b) $\int \frac{1}{x^2 + x} \, dx$

Vis utregningene dine i denne boksen

Task 2

Determine t such that

$$\int_1^t \frac{1}{x+1} \, dx = \ln\left(\frac{5}{2}\right)$$

Task 3

The functions f and g are given by $f(x) = 4 - x^2$ and $g(x) = x + 2$. The graphs of f and g bound a region. What is the area of this region?

Task 4

The sum of the first ten terms of an arithmetic sequence is 90.

Determine the common difference d , given that $a_1 = 3$.

Task 5

The points $A(1, 2, 0)$, $B(4, 6, 2)$, and $C(8, 6, 4)$ are given.

- a) Determine the area of $\triangle ABC$.
- b) Find the distance from point C to the line through A and B .

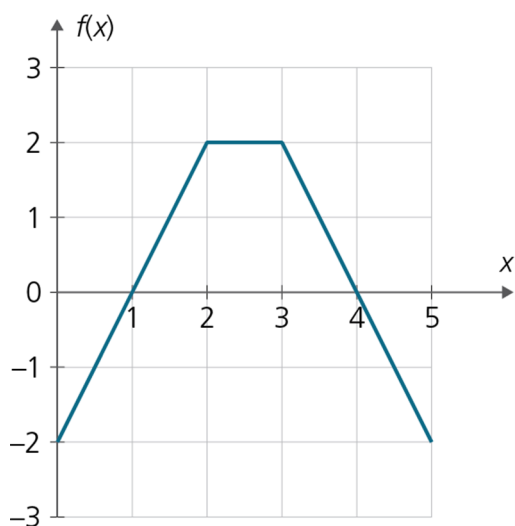
Task 6

Given the planes $\alpha : x + y + z = 6$ and $\beta : 2x - y + z = 3$.

Find a parametric equation for the line of intersection between the two planes.

Task 7

The figure below shows the graph of the function f .



The function g is given by $g(x) = \int_0^x f(t) dt$.

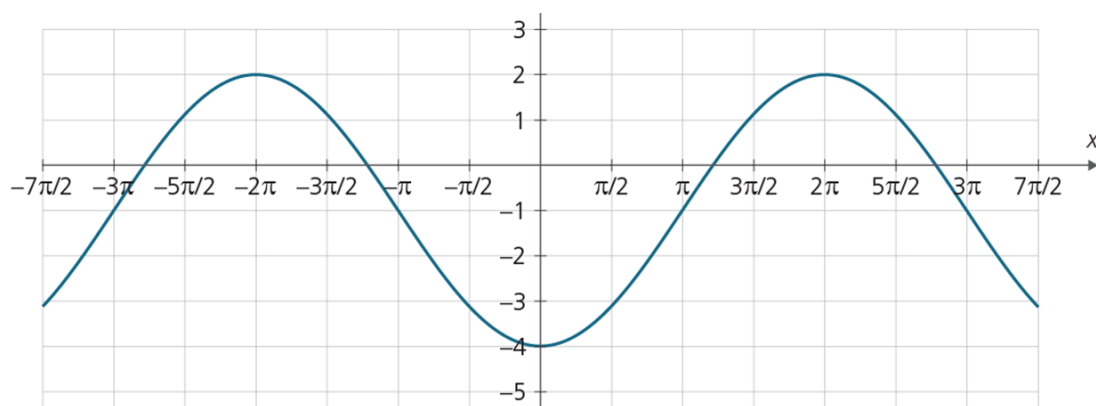
Determine $g(4)$ and $g'(4)$.

Task 8

An angle θ in the second quadrant is given by $\sin \theta = \frac{3}{5}$.

- a) Determine $\cos \theta$ and $\tan \theta$.
- b) Solve the equation $2 \cos(2x) = 1$ for $x \in [0, 2\pi]$.

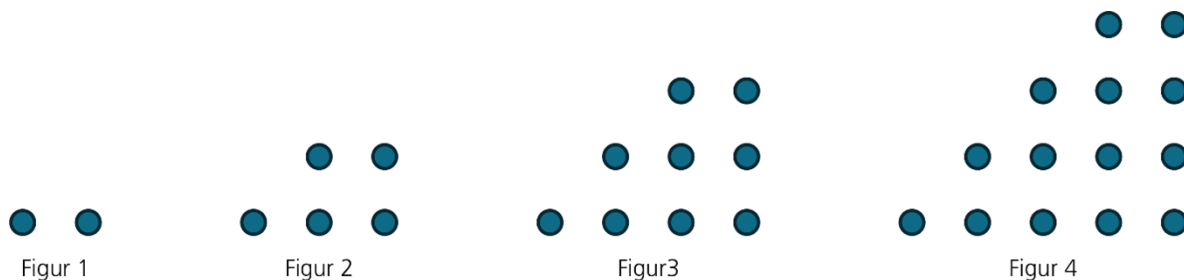
The figure below shows the graph of a function f .



c) Find an expression for f .

Task 9

Each figure below consists of dots arranged in a specific pattern.



Let F_n be the number of dots in figure n . The first four figure numbers are 2, 5, 9, and 14.

- Determine F_6 .
- It can be shown that $F_n = F_{n-1} + n + 1$. Carry out an induction proof showing that $F_n = \frac{1}{2}n^2 + \frac{3}{2}n$ for all $n \geq 1$.

Task 10

Aisha is saving for a down payment to buy a home. On the 1st of each month, she transfers NOK 12,000 to a savings account with 0.3% monthly interest. The first transfer was on 1 January 2023.

- Explain why the very first deposit has been in the account for 36 months by the end of December 2025.
- How much does she have in the account by the end of December 2025?
- How much would she need to deposit each month for the account balance to be half a million NOK by the end of December 2025?